

ALVA's INSTITUTE OF ENGINEERING & TECHNOLOGY, MOODBIDRI

(Autonomous Institution, Affiliated to VTU, Belagavi)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(Accredited by NBA New Delhi (2019-2028) & NAAC with A+ Grade)

Bachelor of Engineering (B.E.)

COMPUTER SCIENCE & ENGINEERING

**Autonomous 2025 Scheme of Teaching and Examinations
&
III to VIII Semester**

Outcome Based Education (OBE) and Choice Based Credit System (CBCS)

(As per Autonomous Regulations of VTU)

(w.e.f A.Y 2026-27)

III SEMESTER-Autonomous Scheme of Teaching and Examinations for A.Y 2026-27

Sl.	Course Code	Course Title	Category	Teaching Dept.	Teaching Hours/Week			Examination (Marks)				Credits
					L	T	P	Hrs	CIE	SEE	Total	
1	1BCS301	Probability, Distributions and Statistics	ASC	Maths	3	2	0	3	50	50 [#]	100	4
2	1BCS302	Object Oriented Programming with Java	IPCC	CSE	3	0	2	3	50	50 [#]	100	4
3	1BCS303	Digital Design and Computer Organization	IPCC	CSE	3	0	2	3	50	50 [#]	100	4
4	1BCS304	Operating Systems	PCC	CSE	3	0	0	3	50	50 [#]	100	3
5	1BCS305	Data Structures and Applications	PCC	CSE	3	0	0	3	50	50 [#]	100	3
6	1BCSL306	Data Structures Laboratory	PCCL	CSE	0	0	2	3	50	50 [#]	100	1
7	1BCSL307	Project Management with Git	AEC	CSE	0	0	2	3	50	50 [#]	100	1
8	1BCP308	Community Project (Project-Based Learning) / Societal Project	SDC	CSE	0	0	2	3	50	50 [#]	100	1
9	1BNSS309	National Service Scheme (NSS)	MC	NSS Coordinator	0	0	2	-	100	-	100	PP
	1BPE309	Physical Education(PE)(Sports and Athletics)		Physical Education								
	1BYOG309	Yoga		Yoga Teacher								
	1BMUK309	Music		Music Teacher								
10	1BESD310	Employability Skill Development -I	SDC	Any	0	2	0	-	100	-	100	PP
Total Credits												21

1BMATDIP311	Mathematics-I for CSE Lateral Entry	MC	Maths / Online	1	0	0	-	100	-	100	PP
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IV SEMESTER- Autonomous Scheme of Teaching and Examinations for A.Y 2026-27

Sl.	Course Code	Course Title	Category	Teaching Dept.	Teaching Hours/Week			Examination Marks)				Credits
					L	T	P	Hrs	CIE	SEE	Total	
1	1BCS401	Discrete Mathematics and Graph Theory	ASC	CSE/Maths	3	2	0	3	50	50 [#]	100	4
2	1BCS402	Database Management Systems	IPCC	CSE	3	0	2	3	50	50 [#]	100	4
3	1BCS403	Computer Networks	IPCC	CSE	3	0	2	3	50	50 [#]	100	4
4	1BCS404	Analysis and Design of Algorithms	PCC	CSE	2	2	0	3	50	50 [#]	100	3
5	1BCSL405	Analysis and Design of Algorithms Lab	PCCL	CSE	0	0	2	3	50	50 [#]	100	1
6	1BCS406x	Ability Enhancement Course	AEC	CSE	0	0	2	3	50	50 [#]	100	1
7	1BCS407	Biology for Computer Engineers	ASC	CHE	2	0	0	2	50	50 [#]	100	2
8	1BEP408	Environmental Science Project	SDC	CSE	0	0	2	2	50	50 [#]	100	1
9	1BNSS309	National Service Scheme (NSS)	MC	NSS Coordinator	0	0	2	-	100	-	100	PP
	1BPE309	Physical Education(PE)(Sports and Athletics)		Physical Education								
	1BYOG309	Yoga		Yoga Teacher								
	1BMUK309	Music		Music Teacher								
10	1BESD410	Employability Skill Development-II	SEC	Any	0	2	0	-	50	50	100	1
Total Credits												21

1BMATDIP411	Mathematics-II for CSE Lateral Entry	MC	Maths / Online	1	0	0	-	100	-	100	PP
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[#]SEE is to be conducted for 100 marks & Scaled down to 50 marks

Ability Enhancement Course(AEC)	
1BCSL406A	UI/UX Design with iOS
1BCSL406B	Elements of Cyber Security
1BAIL406C	Mongo DB
1BAIL406D	Python for Artificial Intelligence
1BCSL406E	Unix System Programming

Audit Courses/Add-on Courses(Online)- 2nd Year (3rd & 4th Sem)

Credits	0	Mode of Study & Completion	Online
Attendance Requirement	Yes	Assessment	Online & Offline
Pass grade	AU	CGPA	No weightage

Course Code	Course Name	University / Platform	Link for online course
1BCSAD401	Salesforce Developer Catalyst	Salesforce AI - Agentforce	https://trailhead.salesforce.com/users/srebello7/trailmixes/salesforce-ai-with-agnetblazer-2026
1BCSAD402	Vibe Coding Essentials - Build Apps with AI Specialization	Scrimba	https://www.coursera.org/specializations/vibe-coding
1BCSAD403	MongoDB Administration Certification	Infosys Spring Board	https://infyspringboard.onwingspan.com/web/en/login
1BCSAD405	Open Source Programming - Master's	Coursera-Illinois Tech	https://www.coursera.org/learn/illinois-tech-open-source-programming-mit
1BCSAD406	Foundational Mathematics for AI	Coursera-Johns Hopkins University	https://www.coursera.org/learn/foundational-mathematics-for-ai
1BCSAD407	DevOps, DataOps, MLOps	Coursera-Duke University	https://www.coursera.org/learn/devops-dataops-mlops-duke
1BCSAD408	Health Data Science Foundation	Coursera-University of Illinois Urbana-Champaign	https://www.coursera.org/learn/health-data-science-foundation

V SEMESTER-Autonomous Scheme of Teaching and Examinations for A.Y 2027-28

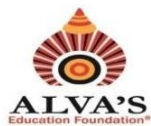
Sl.	Course Code	Course Title	Category	Teaching Dept.	Teaching Hours/Week			Examination (Marks)				Credits
					L	T	P	Hrs	CIE	SEE	Total	
1	1BCS501	Software Engineering and Project Management	HSC	CSE	3	0	0	3	50	50 [#]	100	3
2	1BCS502	Advanced Java Programming	IPCC	CSE	3	0	2	3	50	50 [#]	100	4
3	1BCS503	Theory of Computation	IPCC	CSE	3	0	2	3	50	50 [#]	100	4
4	1BCS504	Computer Vision	PCC	CSE	3	0	0	3	50	50 [#]	100	3
5	1BXX505x	Professional Elective Course-I	PEC	CSE	3	0	0	3	50	50 [#]	100	3
6	1BRM506	Research Methodology and IPR	AEC	CSE	2	0	0	3	50	50 [#]	100	2
7	1BCSL507	Enterprise Application Development with Spring Boot Lab	PCCL	CSE	0	0	2	3	50	50 [#]	100	1
8	1BCS508	Hackathon-Based Project	SDC	CSE	0	0	2	3	50	50 [#]	100	2
9	1BESD509	Soft Skills & Professional Ethics	SDC	Any	0	2	0	-	100	-	100	PP
10	1BESD510	Employability Skill Development -III	SDC	Any	0	2	0	-	100	-	100	PP
Total Credits												22

Professional Elective Course -I			
1BCD505A	Artificial Intelligence	1BCS505D	No SQL Databases
1BCS505B	Data Security and Privacy	1BCS505E	Cloud Computing
1BCG505C	Computer Graphics & Visualization		

VI SEMESTER- Autonomous Scheme of Teaching and Examinations for A.Y 2027-28

Sl .	Course Code	Course Title	Category	Teaching Dept.	Teaching Hours/Week			Examination (Marks)				Credits
					L	T	P	Hrs	CIE	SEE	Total	
1	1BCS601	Machine Learning	IPCC	CSE	3	0	2	3	50	50 [#]	100	4
2	1BCS602	Cryptography and Network Security	PCC	CSE	3	0	0	3	50	50 [#]	100	3
3	1BCS603	Compiler Design	PCC	CSE	3	0	0	3	50	50 [#]	100	3
4	1BCS604	IoT Applications and Automation	IPCC	CSE	3	0	0	3	50	50 [#]	100	3
5	1BXX605x	Professional Elective Courses-II	PEC	CSE	3	0	0	3	50	50 [#]	100	3
	1BCS606	Enterprise Application Development with Spring Boot Lab	PCCL	CSE	0	0	2	3	50	50 [#]	100	1
6	1BXXL607x	Ability Enhancement Course	AEC	CSE	0	0	2	3	50	50 [#]	100	1
7	1BCS608	Capstone Project - Phase I	SDC	CSE	0	0	6	-	100	-	100	3
8	1Bxx609	Universal Human Value	SDC	CSE	1	0	0	-	100	-	100	PP
9	1BESD610	Employability Skill Development -IV	SDC	Any	0	2	0	-	100	-	100	PP
Total Credits											21	

Professional Elective Course -II		Ability Enhancement Courses(AEC)	
1BCS605A	Prompt Engineering	1BAIL607A	Generative Artificial Intelligence
1BCS605B	Blockchain Technology	1BCSL607B	Devops
1BCS605C	Image Processing	1BCSL607C	Angular JS and React
1BCS605D	Big Data Analytics	1BCS607D	Exploratory Data Analysis
1BCS605E	High Performance Computing	1BCS607E	Embedded Secure Element



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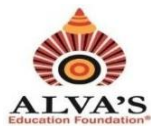
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VII SEMESTER- Autonomous Scheme of Teaching and Examinations for A.Y 2028-29

Sl.	Course Code	Course Title	Category	Teaching Dept.	Teaching Hours/Week			Examination (Marks)				Credits
					L	T	P	Hrs	CIE	SEE	Total	
1	1BCS701	Distributed Computing	IPCC	CSE	3	0	2	3	50	50 [#]	100	4
2	1BXX702x	Professional Elective Course-III	PEC	CSE	3	0	0	3	50	50 [#]	100	3
3	1BXX703x	Professional Elective Course -IV	PEC	CSE	3	0	0	3	50	50 [#]	100	3
4	1BXX704x	Open Elective Course-I	OEC	CSE	3	0	0	3	50	50 [#]	100	3
5	1BCS705	Capstone Project - Phase-II	SDC	CSE	0	0	14	3	100	100	200	7
6	1BIKS706	Indian Knowledge System	MC	Any	1	0	0	-	100	-	100	PP
Total Credits												20

Professional Elective Course -III		Professional Elective Course -IV		Open Elective Course-I	
1BCS702A	AI of things	1BAI703A	Agentic Artificial Intelligence	1BCS704A	Introduction to Data Structures
1BCS702B	Ethical Hacking	1BCS703B	Secure Computation	1BCS704B	Introduction to Database Systems
1BCS702C	Virtual Reality	1BCS703C	Robotic Process Automation with UiPath	1BCS704C	Introduction to Algorithms
1BCS702D	Social and Web Analytics	1BCS703D	Quantum Computing	1BXX704D	Foreign Language (NPTEL/SWAYAM/online VTU)
1BCS702E	Parallel Computing	1BCS703E	Data Engineering and MLOps		



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VIII SEMESTER- Autonomous Scheme of Teaching and Examinations for A.Y 2028-29

Sl.	Course Code	Course Title	Category	Teaching Dept.	Teaching Hours/Week			Examination (Marks)				Credits
					L	T	P	Hrs	CIE	SEE	Total	
1	1BCS801x	Professional Elective-V (NPTEL Course)	PEC	CSE	3	0	0	3	50	50 [#]	100	3
2	1BCS802x	Open Elective-II (NPTEL Course)	OEC	CSE	3	0	0	3	50	50 [#]	100	3
3	1BCS803x	Internship (15 weeks or 90 working days)	SDC	CSE	0	0	24	3	100	100	200	9
Total Credits												15

Professional Elective -V		Open Elective -II		Internship	
1BCS801A	Human-Centered AI	1BCS802A	Cloud Computing	1BCS803A	Industry Internship
1BCS801B	Industrial Cyber Security	1BCS802B	Generative AI	1BCS803B	Research Internship
1BCG801C	Animation Principles and Design	1BCS802C	Prompt Engineering	1BCS803C	Post-Placement Internship
1BCS801D	Scalable Data Systems	1BCS802D	Blockchain Technologies	1BCS803D	Societal Internship
BoS may suggest the latest Courses which is available in NPTEL					